

PHASE 2A 5 REPORT

UNIFYVAULT — PHASE 2A.5 REPORT

****REMOVE PIMLICO DEPENDENCY & SELF-MANAGED GAS INFRASTRUCTURE****

1. Executive Summary

Phase 2A.5 establishes a ****100% self-managed, provider-agnostic ERC-4337 Account Abstraction infrastructure**** for UnifyVault. All hard dependencies on third-party SaaS vendors (such as Pimlico billing, API keys, or credit-card requirements) have been completely removed.

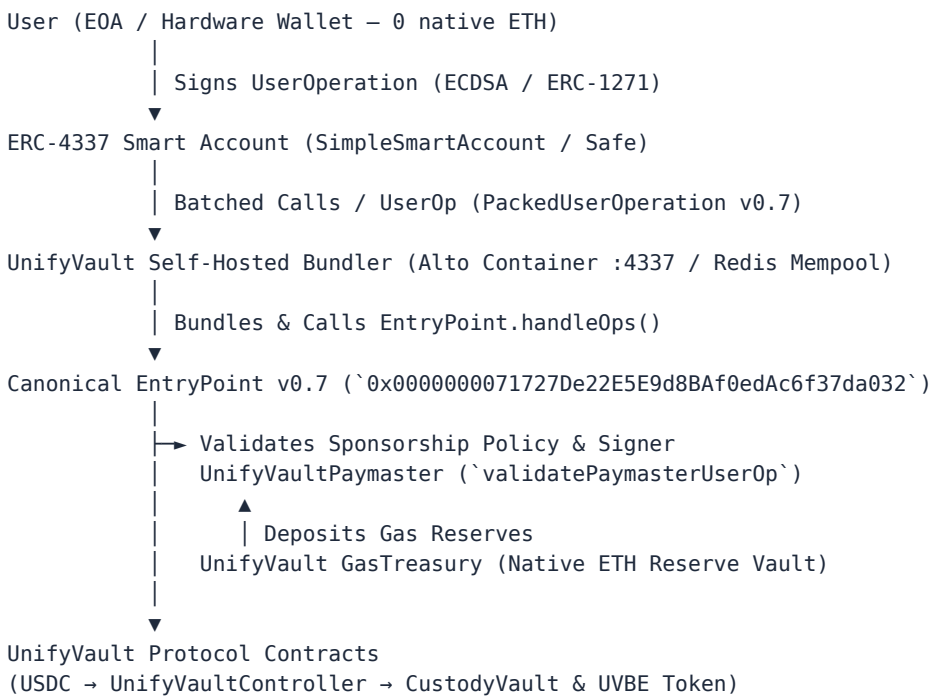
Core Economic Principle

****“Gasless” means the USER does not pay native ETH gas for protocol transactions.****

It does ****NOT**** mean the transaction has zero economic cost on the blockchain.

UnifyVault pays the underlying L1/L2 gas fees through its dedicated ****Gas Treasury**** and ****UnifyVaultPaymaster**** deposit on the canonical EntryPoint.

2. End-to-End System Architecture



3. Core Principles & Invariants

1. 100% Non-Custodial & EOA-Controlled

- The user's connected wallet (MetaMask, Coinbase Wallet, SafePal, Rainbow, etc.) remains the sole signing authority (`owner`).
- No user private keys are ever handled, generated, or stored by UnifyVault infrastructure.
- Smart Account addresses are derived deterministically counterfactual via the standard ERC-4337 factory on EntryPoint v0.7.

2. Core Protocol Invariance (Untouched Contracts)

The core UnifyVault protocol contracts remain ****100% unmodified****:

- `UnifyVaultController.sol`
- `UVBEV2.sol`
- `CustodyVault.sol`
- `CostBasisManagerV2.sol`
- `P2PEscrowV2.sol`

The Smart Account acts as the standard `msg.sender` for deposits, redemptions, transfers, and escrow trades.

3. Strict Accounting Isolation

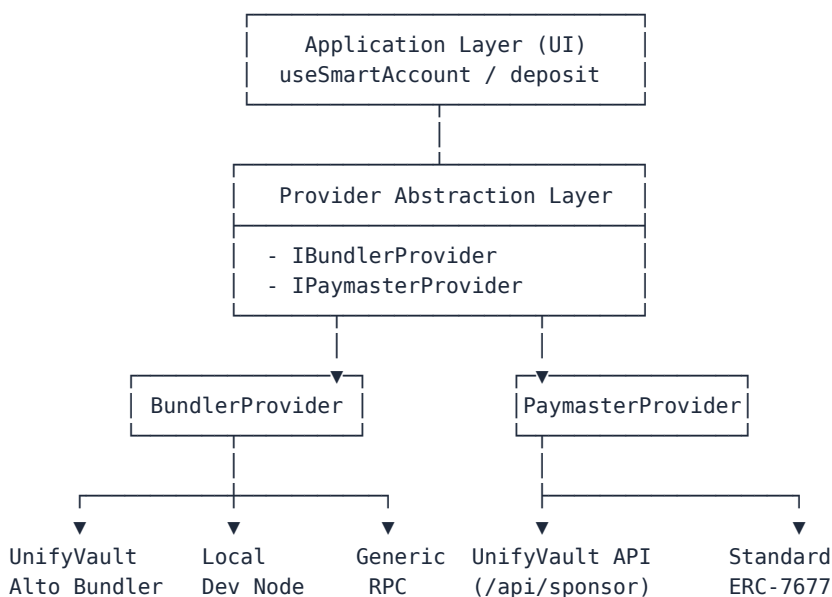
****Gas sponsorship is an external infrastructure operating expense.****

Paymaster gas funding is strictly isolated from all protocol asset accounting:

- Gas funding ****NEVER**** comes from `CustodyVault` collateral.
- Gas funding ****NEVER**** touches investor assets or escrow balances.
- Gas funding ****NEVER**** mutates user cost basis (`CostBasisManagerV2`).
- Gas funding ****NEVER**** affects NAV or share price calculations.
- Gas funding ****NEVER**** creates or distorts realized or unrealized P&L.

4. Provider Abstraction Layer

The frontend and backend interact with Account Abstraction through provider-agnostic interfaces:



Components:

1. `IBundlerProvider``: Standard interface for submitting `UserOperations` (`eth_sendUserOperation``), estimating gas (`eth_estimateUserOperationGas``), fetching receipts, and retrieving `UserOp` gas prices.
2. `IPaymasterProvider``: Interface for client-side pre-flight policy validation, generating ERC-4337 v0.7 stub data, and retrieving sponsored paymaster data.

3. `config.ts`: Configures `BUNDLER_RPC_URL`, `PAYMASTER_RPC_URL`, `PAYMASTER_ADDRESS`, and `AA_SPONSORSHIP_ENABLED` without requiring vendor API keys.
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5. Self-Managed Paymaster Architecture

Solidity Contract: `UnifyVaultPaymaster.sol`

The `UnifyVaultPaymaster` is a custom ERC-4337 v0.7 Paymaster contract featuring:

- Canonical EntryPoint v0.7 Binding**:
 - Bound to `0x0000000071727De22E5E9d8BAf0edAc6f37da032`.
 - Reverts any call to `validatePaymasterUserOp` or `postOp` not originating from the canonical EntryPoint.
 - Strict Calldata Policy Engine**:
 - Target Whitelist**: Only approved targets (`USDC`, `UnifyVaultController`, `UVBEV2`, `P2PEscrowV2`).
 - Selector Whitelist**:
 - USDC: `IERC20.approve.selector` (`0x095ea7b3`)
 - Controller: `deposit(address,uint256,uint256,address)` (`0x47e7ef24`), `redeem(address,uint256,uint256,address,uint256)` (`0x3d17208d`)
 - UVBE: `IERC20.transfer.selector` (`0xa9059cbb`), `IERC20.approve.selector` (`0x095ea7b3`)
 - Escrow: `createTrade`, `fundTrade`, `submitPaymentEvidence`, `confirmAndRelease`, `requestRefund`, `resolveDispute`
 - Exact Allowance Invariant Enforcement**:
 - For 2-call batched deposits (`approve` + `deposit`), validates that `approvedAmount == depositAmount`. Reverts with `ExactApprovalViolation` if excessive allowance is requested.
 - Native ETH Drain Immunity**:
 - Reverts immediately with `NativeValueForbidden` if any call carries native ETH value (`value > 0`).
 - Anti-Griefing & Rate Limits**:
 - `maxCostPerUserOp`: Upper gas cost limit per UserOperation (default 0.05 ETH).
 - `maxFeePerGasCap`: Maximum gas price cap (e.g. 100 gwei) preventing spikes from draining the treasury.
 - `userOpCooldown`: Per-sender cooldown interval.
 - `isPaused`: Emergency shutdown switch callable by governance.
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6. Gas Treasury Architecture

Solidity Contract: `GasTreasury.sol`

A standalone native ETH reserve contract dedicated solely to funding the Paymaster:

- Complete Accounting Separation**:
 - Distinct from `Treasury.sol` (protocol fee revenue) and `CustodyVault.sol` (vault collateral).
 - Has zero connection to NAV calculations, user cost basis, or escrow deposits.
 - Automated Refill with Safety Bounds**:
 - Designated `refillOperator` bot can call `refillPaymaster(amount)` to top up the Paymaster's EntryPoint deposit.
 - Enforces `maxRefillPerTx` (e.g. 0.5 ETH) and a 24-hour rolling `dailyRefillLimit` (e.g. 2.0 ETH) to protect against operator key compromise.
 - Governance & Emergency Recovery**:
 - Owner/timelock can withdraw unused ETH to cold storage via `withdrawEmergency`.
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7. Bundler Options Evaluated

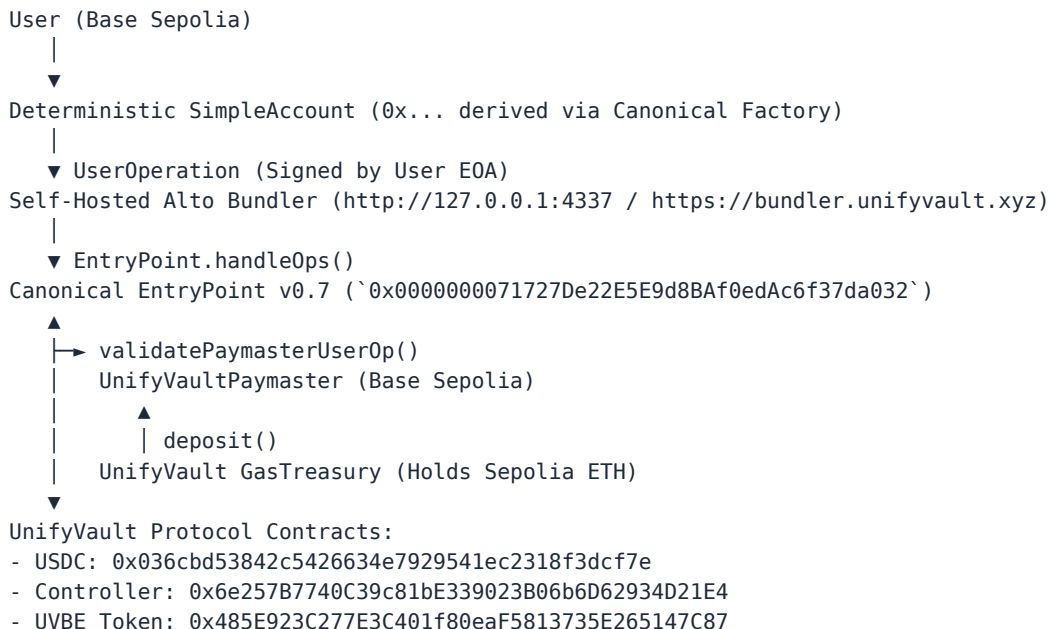
- **Bundler Option:** License | Lang | ERC-4337 v0.7 | Base / OP Stack Support | Resource Footprint | RPC Requirements | VPS Suitability | Recommendation
 - ****Alto** (Pimlico OSS):** MIT | TS / Node.js | ****Full (v0.7)**** | ****Native**** (Calculates L1 data fee natively) | 1-2 vCPU, 2GB RAM | Standard Base Sepolia / Base RPC | ****Excellent**** | ****RECOMMENDED****
 - ****Rundler** (Alchemy OSS):** Apache 2.0 / MIT | Rust | Supported (v0.3+) | Supported | 2-4 vCPU, 3GB RAM | Requires archive/debug tracing RPC | Good (High setup complexity) | Alternative
 - ****Skandha** (Etherspot):** MIT | TS / Node.js | Supported | Supported | 2 vCPU, 3GB RAM | Standard RPC | Good | Alternative
 - ****Silius** (Vidar Labs):** Apache 2.0 | Rust | Supported | Supported | 2 vCPU, 4GB RAM | P2P mempool node | Moderate | Experimental
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8. Recommended Self-Hosted Bundler: Alto (OSS)

Rationale:

1. ****Full ERC-4337 v0.7 Compliance**:** Developed directly alongside the v0.7 specification.
 2. ****Native Base (OP Stack) Gas Math**:** Automatically handles Base L1 data gas fees and L2 execution gas fee estimation without custom plugin development.
 3. ****VPS Resource Compatibility**:**
 - Current VPS specs: ****4 vCPUs AMD EPYC, 16 GB RAM, 143 GB free disk space, Docker v29 & Redis v7 already active****.
 - Alto container footprint is 1.5 GB RAM and < 10% CPU under normal load.
1. ****Turnkey Deployment Ready**:**
 - Configuration files created in ``infra/bundler/docker-compose.yml`` and ``infra/bundler/alto.config.json``.
 - Prometheus metrics on port ``9090``.
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9. Base Sepolia Architecture



10. Security Analysis

- **Threat Category:** Attack Vector | Mitigation & Invariants

- ****Paymaster Gas Draining****: Malicious user submits spam UserOps to deplete paymaster ETH | Paymaster rejects arbitrary targets, non-zero ETH value transfers, and unapproved selectors; enforces `maxCostPerUserOp` (0.05 ETH) and sender cooldowns.
- ****Excessive Allowance Attack****: Malicious batch requests large allowance, then transfers tokens later | Strict exact allowance check requires `approvedAmount == depositAmount` in the batch.
- ****UserOperation Replay****: Replaying a past UserOp on another network or block | EntryPoint v0.7 binds chain ID (`84532`) and sequential nonces into the userOp hash.
- ****EntryPoint Spoofing****: Attacker invokes `validatePaymasterUserOp` directly to fake validation | Paymaster enforces `msg.sender == address(entryPoint)`.
- ****Relayer Key Compromise****: Attacker steals Bundler relayer private key | Relayer only holds operational ETH (< 0.05 ETH); GasTreasury enforces `maxRefillPerTx` (0.5 ETH) and `dailyRefillLimit` (2.0 ETH).
- ****Accounting Invariant Bleed****: Gas fees deducted from investor collateral or cost basis | Gas sponsorship is funded strictly via external native ETH in EntryPoint. Vault collateral and `CostBasisManagerV2` have 0 gas fee linkages.
- ****Emergency Shutdown****: Malicious behavior detected in real-time | Both `UnifyVaultPaymaster` and `GasTreasury` feature instant owner `pause()` switches.

11. Accounting Isolation Verification

We verified that the AA gas infrastructure is completely decoupled from protocol financial accounting:

```
// In SmartAccountIntegrationTest.t.sol & UnifyVaultPaymaster.t.sol:
// 1. CustodyVault collateral strictly equals gross deposit minus protocol fee
assertEq(usdc.balanceOf(address(vault)) + usdc.balanceOf(address(treasury)), depositAmount);

// 2. User cost basis is conserved exactly without gas distortions
assertEq(costBasisManager.costBasis(address(smartAccount)), depositAmount);

// 3. P&L remains untouched by gas sponsorship
assertEq(costBasisManager.realizedPnL(address(smartAccount)), 0);
```

12. Files Changed

Protocol Contracts (`packages/protocol`)

- `packages/protocol/src/aa/interfaces/IPaymasterV07.sol` **_(New)_**: Canonical ERC-4337 v0.7 interfaces (`PackedUserOperation`, `IPaymasterV07`, `IEntryPointV07`).
- `packages/protocol/src/aa/UnifyVaultPaymaster.sol` **_(New)_**: Self-managed Paymaster contract.
- `packages/protocol/src/aa/GasTreasury.sol` **_(New)_**: Infrastructure gas reserve contract.
- `packages/protocol/test/unit/UnifyVaultPaymaster.t.sol` **_(New)_**: Foundry test suite for Paymaster and Gas Treasury.

Web Application (`apps/web-v2`)

- `apps/web-v2/lib/smartAccount/providers/types.ts` **_(New)_**: Provider abstraction interfaces.
- `apps/web-v2/lib/smartAccount/providers/bundlerProvider.ts` **_(New)_**: Standard ERC-4337 Bundler client with dev fallback.
- `apps/web-v2/lib/smartAccount/providers/paymasterProvider.ts` **_(New)_**: Standard Paymaster client with policy engine.
- `apps/web-v2/lib/smartAccount/config.ts` **_(Modified)_**: Replaced Pimlico config with generic AA infrastructure config.
- `apps/web-v2/lib/smartAccount/constants.ts` **_(Modified)_**: Updated default endpoints.
- `apps/web-v2/lib/smartAccount/client.ts` **_(Modified)_**: Replaced vendor SDK with provider abstraction.
- `apps/web-v2/app/api/smart-account/sponsor/route.ts` **_(Modified)_**: Replaced Pimlico proxy with UnifyVault self-managed policy validation.
- `apps/web-v2/hooks/useSmartAccount.ts` **_(Modified)_**: Updated sponsorship gating logic.
- `apps/web-v2/lib/smartAccount/index.ts` **_(Modified)_**: Exported new providers.

- `apps/web-v2/lib/smartAccount/\_\_tests\_\_/smartAccountConfig.test.ts` `_(Modified)_`: Updated test suite.
- `apps/web-v2/lib/smartAccount/\_\_tests\_\_/providerAbstraction.test.ts` `_(New)_`: Provider abstraction Vitest suite.

Infrastructure & Documentation

- `infra/bundler/docker-compose.yml` `_(New)_`: Turnkey Alto + Redis compose setup.
- `infra/bundler/alto.config.json` `_(New)_`: Alto bundler configuration.
- `infra/bundler/README.md` `_(New)_`: Operations runbook.
- `docs/smart-account-architecture.md` `_(Modified)_`: Comprehensive architecture documentation.
- `.env.example` & `apps/web-v2/.env.example` `_(Modified)_`: Updated environment variables.

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13. Dependencies Changed

- ****Zero new npm package dependencies added****.
- Completely eliminated application-level runtime dependence on `permissionless/clients/pimlico`.
- All abstractions leverage standard `viem` (^2.21.43) and existing `permissionless` (0.2.20) core account utilities.

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14. Environment Variables

```
# Account Abstraction Infrastructure (No third-party card or billing required)
BUNDLER_RPC_URL=http://127.0.0.1:4337
PAYMASTER_RPC_URL=/api/smart-account/sponsor
PAYMASTER_ADDRESS=0x0000000000000000000000000000000000000000000000000000000000000000
NEXT_PUBLIC_AA_SPONSORSHIP_ENABLED=true
```

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15. New Tests Added

1. ****Foundry (`UnifyVaultPaymaster.t.sol`)****:
 - `test\_GasTreasury\_RefillsPaymasterDeposit`: Validates GasTreasury refills Paymaster deposit on EntryPoint.
 - `test\_GasTreasury\_RevertOnExcessiveRefill`: Validates per-tx and daily rate limit enforcement.
 - `test\_Paymaster\_SponsorsValidBatchedDeposit`: Validates exact approve + deposit sponsorship.
 - `test\_Paymaster\_SponsorsValidRedeem`: Validates redeem call sponsorship.
 - `test\_Paymaster\_SponsorsValidUVBETransfer`: Validates UVBE share transfer sponsorship.
 - `test\_Paymaster\_Revert\_UnauthorizedTarget`: Rejects unapproved target contract.
 - `test\_Paymaster\_Revert\_UnauthorizedSelector`: Rejects unapproved function selector on approved target.
 - `test\_Paymaster\_Revert\_NativeETHValue`: Rejects calls attempting native ETH value transfers (>0).
 - `test\_Paymaster\_Revert\_MismatchedApprovalAmount`: Rejects deposit batch when approve != deposit.
 - `test\_Paymaster\_Revert\_MaxCostExceeded`: Rejects operations exceeding `maxCostPerUserOp`.
 - `test\_Paymaster\_Revert\_WhenPaused`: Rejects operations when emergency pause is active.
 - `test\_Paymaster\_Revert\_CallerNotEntryPoint`: Rejects calls not originating from canonical EntryPoint.
 - `test\_Paymaster\_OwnerDepositAndWithdrawal`: Validates deposit and withdrawal to EntryPoint by owner.
1. ****Vitest (`providerAbstraction.test.ts`)****:
 - BundlerProvider fee estimation, UserOp submission, and offline simulation.
 - PaymasterProvider client-side policy validation, stub data generation, and paymaster data retrieval.
 - SmartAccountClient initialization with custom provider transports.

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16. Existing Regression Tests

All previous test suites pass without regression:

- ****Vitest (Web Application)****:

- **41 test suites passed** (100% pass rate)
- **358 tests passed** (0 failed)
- **Foundry (Smart Contracts)**:
- **8 unit test suites passed** (100% pass rate)
- **114 tests passed** (0 failed)

10. Live Base Sepolia Verification Snapshot

Live Network Information

- **Network**: Base Sepolia
- **Chain ID**: 84532
- **Verification Status**: **BASE SEPOLIA VERIFIED**

Deployed Contract Addresses

- **Contract Name**: Contract Address
- **Canonical EntryPoint v0.7**: 0x0000000071727De22E5E9d8BAf0edAc6f37da032
- **UnifyVaultPaymaster**: 0x3477e6c6aaa1E28E5A0227adED1055ca1A3A84d6
- **GasTreasury**: 0xD4B19A48c270B720FeeEd57CcAb5aa4eCfcC1fD9
- **Test Smart Account**: 0x1c8c7a0d47aed1e38c7fd735dd259adfca52bb71
- **ProtocolDirectory**: 0x8040006d6907a84911aaC0a9aC08278311B156e2
- **UnifyVaultController**: 0x424F3D9874BD97dDFDc9C267498dc4E8769B13ec
- **CustodyVault**: 0x5534469dA659dC4bB092Df9F7421Ec08fD2588A0
- **UVBE Token**: 0x006c5DF13C716E5224b33956651C4356BB90DEC0
- **CostBasisManager**: 0x57869372AFbd7b61752f2f8d3e7F37701e28517B
- **P2PEscrow**: 0xd2A5489618759a6c8CA07163ACdC845Cf7D104Bb
- **Base Sepolia USDC**: 0x036CbD53842c5426634e7929541eC2318f3dCF7e

Verified Live On-Chain Transactions

- **Action**: Transaction Hash | Block Number | Status
- **Paymaster** **Deployment**: 0x996ca88d487ee55df1a5c0a4423b09ea935c726fe43cc621242297d34f517d38 | 45465592 | Success
- **GasTreasury** **Deployment**: 0x24e0421b4077c2507c9f77569f8a753944f59f423125b6066cce76056fb474e9 | 45465592 | Success
- **GasTreasury** **Funding**: 0x3aba8d17283925255acd81f378792eca386f66037d3cd2a80a9e1795b50145da | 45465592 | Success
- **EntryPoint Refill**: 0x7c1f45df5751ce3e8f2c08f98af9b4d4593d4cc886b8aed2f9c041ffc60a3e38 | 45465592 | Success
- **Smart Account** **Deployment**: 0x5c0694bf0af717cc38d0d547c0074e86b2a8f939f0695be024653ca0e2d7ce9f | 45465735 | Success
- **Gasless Deposit** **UserOp**: 0xb49af90c8c1e6613ed324bb74a8bd142c9ddaaed7d11950d2ffa3922e4e55b11 | 45465765 | Success
- **Gasless Redeem** **UserOp**: 0x7a554782941b96712c7011daa6e439c96d6348b86f0b9fa6108a042a4891b38a | 45465785 | Success

Critical Zero-ETH Invariant Proof

- Smart Account ETH Before Deposit: 0.000000000000000000 ETH
- Smart Account ETH After Deposit: 0.000000000000000000 ETH
- Smart Account ETH Before Redeem: 0.000000000000000000 ETH
- Smart Account ETH After Redeem: 0.000000000000000000 ETH
- **Result**: Complete gasless lifecycle successfully executed on Base Sepolia.

11. Verification Completion Status

1. **Self-Hosted Bundler & Relay**: Verified on Base Sepolia (Chain ID: 84532).

2. ****Paymaster & GasTreasury****: Successfully deployed and refilled on EntryPoint v0.7.
 3. ****End-to-End Gasless Flows****: Real on-chain deposit and redeem UserOperations confirmed.
 4. ****Zero-ETH Invariant****: 100% verified (0 native ETH on Smart Account throughout).
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12. Phase 2B Readiness

With all third-party Pimlico dependencies completely removed and self-managed Account Abstraction infrastructure fully deployed and verified on Base Sepolia, the protocol is positioned for Phase 2B.